Soumyaranjan Dash

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Research Positions

National Solar Observatory (NSO), Makawao, HI, USA
Brinson Prize Postdoctoral Fellow, *Jun 2025 – Present*
Mentor: Dr. Thomas A. Schad

- Lead the analysis of coronal spectropolarimetry from DKIST/Cryo-NIRSP. Wrote `cn-specfit` and `cn-polfit`, pipelines that fit every Cryo-NIRSP spectrum and its polarization to map coronal density, temperature, flows and magnetic topology.
- Forward-model Fe XIII polarized emission from global and regional coronal models (POT3D, AMRVAC, pyCELP) and compare it with Cryo-NIRSP, including coordinated near-quadrature observations with Solar Orbiter.
- Wrote SFT2D, an open, flux-conserving surface flux transport code. Calibrated once, it reproduces more than a century of polar-field reversals.

Institute for Astronomy, University of Hawai'i at Mānoa, Pukalani, HI, USA
Solar Physics Postdoctoral Fellow, *Mar 2023 – Jun 2025*
Mentor: Dr. Xudong Sun

- Built an Ensemble Kalman Filter framework that utilizes a surface flux transport model to infer the Sun's surface flows, and tested it with an observing system simulation experiment.
- Led the study within a collaboration of IfA, NSO, LMSAL, NCAR/HAO and Stanford, from concept to publication (*ApJ*, 2024).

Education

Indian Institute of Science Education and Research (IISER) Kolkata, West Bengal, India
Doctor of Philosophy (PhD), Center of Excellence in Space Sciences India, *Jul 2016 – Nov 2022*
Supervisor: Prof. Dibyendu Nandy **CGPA: 8.5/10**

Central University of Himachal Pradesh, Dharamshala, India
Master of Science in Theoretical Physics, *Jul 2013 – Jun 2015*
MSc Thesis Supervisor: Dr. Ayan Chatterjee **CGPA: 5.07/6**

Research Interests

- **Coronal magnetometry.** Inferring coronal magnetic fields, density and flows from forbidden-line spectropolarimetry (DKIST/Cryo-NIRSP, UCoMP, Aditya-L1/VELC) through forward synthesis and inversion.
- **Coronal and heliospheric modeling.** Potential-field, magnetofrictional and MHD models of the global and active-region corona (PLUTO, SWMF/BATS-R-US, AMRVAC, POT3D), coronal null points, and predictions of the coronal magnetic field structure for total solar eclipses.
- **Surface flux transport and data assimilation.** Using Ensemble Kalman filtering with flux transport models to infer surface flows, the polar-field precursor, and solar activity prediction.
- **Long-term solar variability.** The Sun's open flux and cosmic-ray modulation through grand minima, maxima and regular activity phases.
- **Physics-guided machine learning.** Learned closures, fast emulators and simulation-based inference that keep the conservation laws of the underlying physics exact.

Refereed Publications

1. **Dash, S.**, DeRosa, M. L., Dikpati, M., Sun, X., Mahajan, S. S., Liu, Y., and Hoeksema, J. T., "Ensemble Kalman Filter Data Assimilation into the Surface Flux Transport Model to Infer Surface Flows: An Observing System Simulation Experiment", 2024, *The Astrophysical Journal*, 975, 288
2. **Dash, S.**, Nandy, D., and Usoskin, I., "Long-term forcing of the Sun's coronal field, open flux, and cosmic ray modulation potential during grand minima, maxima, and regular activity phases by the solar dynamo mechanism", 2023, *Monthly Notices of the Royal Astronomical Society*, 525, 4801
3. Nandy, D., Baruah, Y., Bhowmik, P., **Dash, S.**, Gupta, S., Hazra, S., Lekshmi, B., Pal, S.,

Pal, S., Roy, S., Saha, C., and Sinha, S., "Causality in heliophysics: magnetic fields as a bridge between the Sun's interior and the Earth's space environment", 2023, *Journal of Atmospheric and Solar-Terrestrial Physics*, 248, 106081

4. Nandy, D., Martens, P. C. H., Obridko, V., **Dash, S.**, and Georgieva, K., "Solar evolution and extrema: current state of understanding of long-term solar variability and its planetary impacts", 2021, *Progress in Earth and Planetary Science*, 8, 40
5. **Dash, S.**, Bhowmik, P., Athira, B. S., Ghosh, N., and Nandy, D., "Prediction of the Sun's Coronal Magnetic Field and Forward-modeled Polarization Characteristics for the 2019 July 2 Total Solar Eclipse", 2020, *The Astrophysical Journal*, 890, 37
6. Pal, S., **Dash, S.**, and Nandy, D., "Flux Erosion of Magnetic Clouds by Reconnection With the Sun's Open Flux", 2020, *Geophysical Research Letters*, 47, e2019GL086372
7. **Dash, S.**, Bhowmik, P., and Nandy, D., "Prediction of the Sun's Corona for the Total Solar Eclipse on 2019 July 2", 2019, *Research Notes of the American Astronomical Society*, 3, 86
8. Nandy, D., Bhowmik, P., Yeates, A. R., Panda, S., Tarafder, R., and **Dash, S.**, "The Large-scale Coronal Structure of the 2017 August 21 Great American Eclipse: An Assessment of Solar Surface Flux Transport Model Enabled Predictions and Observations", 2018, *The Astrophysical Journal*, 853, 72
9. **Dash, S.**, and Nandy, D., "A Magnetofrictional model for the solar corona", 2018, *Proceedings of the International Astronomical Union*, 13 (S340), 87

Software

- **SFT2D** (Python, doi:10.5281/zenodo.17459135): conservative finite-volume surface flux transport on a pole-to-pole spherical mesh, with super-time-stepping, a test suite and documentation built in CI.
- **cn-specfit** and **cn-polfit** (Python): pipelines that fit DKIST/Cryo-NIRSP Level-1 spectra and Stokes profiles with a forward model of coronal emission, scattered light, telluric absorption and instrument response.
- **MFC3D / MF3D** (modern Fortran, MPI + OpenMP): a 3D magnetofrictional code on a staggered spherical grid with HDF5 output, run on the CU Boulder Alpine cluster.
- **SFT-1D** (Python + Fortran): the 1D flux transport model and Ensemble Kalman Filter behind the 2024 *ApJ* study.
- **HelioField** (doi:10.5281/zenodo.23487231): the coronal field in 3D for any date since 2010, with PFSS field lines, the current sheet and the magnetic connection of the spacecraft fleet.
- **HelioFleet** (doi:10.5281/zenodo.23328187) and the **CryoNIRSP Viewer**: browser tools for spacecraft positions and for public DKIST Cryo-NIRSP datasets.
- **PLUTO global corona module**: data-constrained global MHD of the solar corona, developed during the PhD.

Talks & Posters

2026 *Diagnosing Coronal Magnetic Field Structure and Thermodynamics with DKIST/Cryo-NIRSP and Forward Modeling*, AAS Solar Physics Division Meeting. Abstract.

2024 *Modeling the evolution of solar coronal magnetic fields and its impact on heliosphere*, 42nd Meeting of the Astronomical Society of India (oral), with a poster on the Ensemble Kalman Filter for surface flux transport.

2023 *Demonstration of Ensemble Kalman Filter Data Assimilation Technique and its Application to Infer the Surface Flows with Surface Flux Transport Simulations*, AGU Fall Meeting (oral), 11 Dec 2023. Abstract.

2023 *Demonstration of Ensemble Kalman Filter Data Assimilation Technique for Application to Surface-Flux Transport Modeling*, 54th Meeting of the AAS Solar Physics Division.

2023 *Forcing of the state of the heliosphere during grand minima, maxima and regular activity phases by the solar dynamo mechanism*, SHINE 2023 Workshop.

2022 *Understanding The Global Coronal Magnetic Field Evolution Using Data-constrained Magnetohydrodynamic Model*, 15th Quadrennial Solar-Terrestrial Physics Symposium (STP-15, oral), 22 Feb 2022.

2021 *Understanding the Global Coronal Magnetic Fields using Data-constrained Magnetohydrodynamic Model*, AGU Fall Meeting (poster), 13 Dec 2021. Poster.

2021 *Data Driven Models for Coronal Magnetic Field: Aditya-L1 Perspective*, workshop colloquium,

Multi-payload and Multi-Observatory Science with Aditya-L1, 39th Meeting of the ASI, 19 Feb 2021.

2020 *Dynamics of the Sun's Polar Field: Possible Insights from Out of the Ecliptic Observations*, AGU Fall Meeting (oral), 4 Dec 2020. Presentation.

Observing Proposals

DKIST Operations Commissioning Phase, Cycle 3 *PI: Dr. Soumyaranjan Dash*
Understanding Physical Mechanisms that Drive Solar Wind with DKIST Coronal Magnetometry. Interchange reconnection at the boundary between open and closed fields should leave strong current sheets, enhanced temperatures and confined upflows. The proposal uses Cryo-NIRSP to map magnetic topology, temperature, electron density and Doppler shifts at such boundaries, to test it against wave-driven solar wind formation.

Awards & Service

Fellowships: Brinson Prize Postdoctoral Fellowship, NSO (2025); INSPIRE Doctoral Fellowship, Department of Science and Technology, Government of India; All India Rank 98, Joint Entrance Screening Test (JEST) 2016.
Peer review: Referee for *The Astrophysical Journal*, *Monthly Notices of the Royal Astronomical Society* and *Open Research Europe*; panel reviewer, NSF Geospace SHINE (2024).
In the news: CESSI team accurately predicts shape of Sun's corona 43 days ahead of eclipse (The Hindu); Total Solar Eclipse 2017: Team led by Indian astrophysicist predicts solar corona (Indian Express); Great American solar eclipse: studying the Sun's corona (The Hindu).

Outreach Projects

Space weather assessment and predictions, CESSI *2017–2023*
PI: Prof. Dibyendu Nandy
A public dashboard, cessi.in/spaceweather, combining satellite and ground-based observations with models of the surface and coronal magnetic field, active-region flare potential and solar wind parameters.

Modelling the progression of COVID-19 in India, CESSI *2020–2022*
PI: Prof. Dibyendu Nandy
An epidemiological model of COVID-19 in India, constrained by parameter-space studies of the national data and published at cessi.in/coronavirus.

Teaching

Teaching assistant at IISER Kolkata: **SS4203** Fluid- and Magnetohydrodynamics; **PH4102** Introductory Astrophysics; **PH3105** Computational Physics.

Computer Skills

Languages: Python, modern Fortran, C, JavaScript, IDL, Bash, LaTeX, HTML/CSS
Scientific Python: NumPy, SciPy, Astropy, SunPy, Matplotlib; packaging, `pytest`, Sphinx
HPC: MPI, OpenMP, SLURM; Linux (Ubuntu, RHEL, CentOS) and macOS
Simulation codes: PLUTO, SWMF/BATS-R-US, MPI-AMRVAC, POT3D, pyCELP
Practice: Git/GitHub, GitHub Actions CI, Zenodo archiving, FITS and HDF5 I/O
Visualization: ParaView, VisIt, Tecplot, MATLAB, three.js

References

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